

Maryam Jabbari

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🌐 Portfolio

Professional Summary

M.Sc. student in Bioinformatics with a B.Sc. in Computer Science and research experience in retrieval-augmented learning, knowledge distillation, and efficient 3D medical image analysis. My current work spans retrieval-augmented protein language models and efficient deep learning for volumetric medical image segmentation. I am particularly interested in multimodal representation learning, foundation models, and efficient deep learning for biomedical data.

Education

2025–Present	M.Sc. in Bioinformatics <i>Amirkabir University of Technology (Tehran Polytechnic), Iran</i> • Thesis: Retrieval-Augmented Protein Language Models for Multi-Label Antimicrobial Peptide Classification • Selected Courses: Analysis of Protein and Nucleic Acid Sequences (20/20) • Machine Learning (19.75/20) • Modeling and Prediction of Macromolecular Structures (19.5/20) • Special Topics: Deep Learning and Computational Physics (19.25/20) GPA: 19.26/20
2021–2025	B.Sc. in Computer Science <i>Amirkabir University of Technology (Tehran Polytechnic), Iran</i> • Thesis: Feature-Based Knowledge Distillation with CBAM Attention Grade: 20/20 Supervisor: Dr. F. Shakeri • Selected Courses: Data Mining (20/20) • Design and Analysis of Algorithms (20/20) • Linear Optimization (20/20) • Fundamental of Bioinformatics (18.92/20) • Neural Networks (18.83/20) • Artificial Intelligence (18.75/20) GPA: 18.87/20

Research Interests

- Multimodal & Representation Learning
- Foundation Models & Retrieval-Augmented Learning
- Efficient Deep Learning & Model Adaptation
- 3D Medical Image Analysis

Research Experience

Jun 2026–Present	Retrieval-Augmented Protein Language Models for Multi-Label Antimicrobial Peptide Classification <i>M.Sc. Thesis under <u>Dr. F. Zare</u> and <u>Dr. Z. Ghorbanali</u>, Amirkabir University of Technology</i> • Adapting a retrieval-augmented protein language model to peptide sequences for multi-label antimicrobial function classification, with comparisons against non-retrieval and frozen-retrieval baselines. • Designing controlled experiments and systematic evaluation protocols to quantify how retrieved sequence information and retrieval configuration affect downstream classification performance.
Sep 2024–Present	Efficiency-Driven Redesign of Convolutional Neural Networks for Medical Image Segmentation <i>Research Project under <u>Dr. F. Shakeri</u> Iran National Science Foundation (INSF) Manuscript in Preparation</i> • Contributing to the design and evaluation of a MobileNetV4-inspired 3D CNN with inverted bottleneck blocks within an nnU-Net-derived framework for efficient volumetric medical image segmentation. • Investigating input-adaptive dynamic convolutions and joint target-spacing/stem-stride planning, and evaluating architectural choices through segmentation quality, runtime, and memory trade-offs.
Jun 2024–May 2025	Feature-Based Knowledge Distillation with CBAM Attention <i>B.Sc. Thesis under <u>Dr. F. Shakeri</u>, Amirkabir University of Technology</i> • Developed a teacher–student knowledge-distillation framework on CIFAR-100 using a ResNet-56 teacher and ResNet-20 student, with CBAM-guided intermediate feature matching. • Combined feature-level transfer with conventional logit distillation and evaluated the effect of attention-guided knowledge transfer on student-model performance. Code
Oct 2022–Dec 2023	Autonomous Vehicle Perception in Simulation <i>Robotics Team Member, Robotics Lab, Amirkabir University of Technology</i> • Developed and tested computer-vision components in the AVIS Engine simulator using Python, OpenCV, NumPy, and YOLO-based obstacle detection. • Contributed to the team awarded 1st Place in the Simulation League at the IRAN FIRA RoboWorld Cup Open.

Selected Projects

Jul 2026–Aug 2026

Multilabel Antimicrobial Peptide Classification Benchmark

- Built a controlled five-label AMP benchmark on ESCAPE using the reciprocal Fold1/Fold2 training protocol and an independent test set.
- Compared seven frozen protein/peptide language-model encoders with six biological sequence representations under a shared flat MLP; also evaluated the biological representations with Random Forest classifiers.
- Implemented overlap auditing, leakage-controlled splits, embedding caching, reciprocal-fold ensembling, and evaluation under severe multilabel imbalance using mAP, macro/micro F1, balanced accuracy, and per-label diagnostics.

Stack: PyTorch · Transformers · scikit-learn · NumPy

Feb 2025–Jun 2025

[Code](#)

Image Reconstruction via Regularized Inverse Problems

- Modeled image degradation as the linear inverse problem $b = Ax + \eta$ using Gaussian blur, motion blur, and additive noise, with the spatial forward operator represented as a SciPy LinearOperator.
- Compared unregularized LSQR, zero-order Tikhonov regularization, Wiener filtering, and Richardson–Lucy deconvolution.
- Analyzed reconstruction quality and parameter sensitivity using PSNR and SSIM with recorded parameter sweeps.

Stack: NumPy · SciPy · scikit-image · Numerical Optimization

[Code](#)

Dimensionality Reduction for scRNA-seq Data

- Built dataset-specific preprocessing pipelines for Deng (GSE45719), Chu (GSE75748), and a filtered PBMC68k subset, including filtering, normalization, log transformation, HVG selection, and feature scaling.
- Compared PCA, t-SNE, UMAP, and a project-specific PyTorch VAE using low-dimensional visualizations and higher-dimensional representations for downstream clustering.
- Evaluated K-means clustering using ARI, AMI, and Silhouette Score and documented methodological differences from the reference benchmark.

Stack: PyTorch · scikit-learn · UMAP · Scanpy/scVelo

Feb 2024–Jun 2024

[Code](#)

Kolmogorov–Arnold Network

- Implemented a KAN from scratch in PyTorch where each connection learns a univariate function using cubic B-spline bases and a learnable SiLU residual component.
- Implemented cubic B-spline evaluation with iterative Cox–de Boor dynamic programming, trainable basis coefficients, and knot grids registered as PyTorch buffers.
- Benchmarked against width- and parameter-matched MLP baselines and added correctness, gradient-flow, and learned edge-function visualization tests.

Stack: PyTorch · NumPy · scikit-learn · B-Splines

[Code](#)

Autograd Engine and Neural Network from Scratch

- Implemented a scalar reverse-mode automatic differentiation engine with computational-graph construction, differentiable scalar operations, topological backpropagation, and gradient accumulation.
- Built Neuron, Layer, and MLP abstractions on the custom engine, including nonlinear activations, MSE loss, gradient-descent updates, and graph visualization.

Stack: Python · Reverse-Mode Autodiff · Backpropagation · Graphviz

Honors & Awards

2026	Ranked 1st in the M.Sc. entering cohort based on cumulative GPA, Amirkabir University of Technology.
2024	Direct admission to the M.Sc. program through the Exceptional Talents pathway , without the national graduate entrance examination and with tuition-free admission. Ranked in the top 10% of the B.Sc. entering cohort based on final cumulative GPA, Amirkabir University of Technology.
2023	1st Place, Simulation League – IRAN FIRA RoboWorld Cup Open , Robotics Team MCAC, Amirkabir University of Technology.

Teaching Assistant Experience

Sep 2026–Present	Bioinformatics (M.Sc.) <u>Dr. F. Zare</u> Preparing and refining lecture materials and contributing to graduate-level assignments and the course project.
Sep 2025–Jan 2026	Deep Learning (B.Sc.) <u>Dr. F. Shakeri</u> - Contributed to preparing and refining course materials and evaluating project presentations, reports, code, and final deliverables. Numerical Linear Algebra (B.Sc.) <u>Dr. M. Dehghan</u> - Contributed to homework/final-project design and evaluation; led the exam-question design subgroup and reviewed questions and solution keys.
Feb 2025–Jan 2026	Artificial Intelligence and Workshop (B.Sc.) <u>Prof. M. Ghatee</u> Contributed to assignment and course-project design and evaluated student presentations and project deliverables.
Feb 2025–Jun 2025	Foundations of Numerical Analysis (B.Sc.) <u>Dr. F. Shakeri</u> - Contributed to problem-set design and evaluated student submissions. Operating Systems (B.Sc.) <u>Dr. Z. Ghorbanali</u> - Prepared course materials, contributed to assignment/project design, and graded student submissions with feedback.
Feb 2024–Jun 2024	Advanced Programming and Workshop (B.Sc.) <u>Prof. M. Ghatee</u> - Contributed to programming exercises/projects and led interactive problem-solving and workshop sessions. Design and Analysis of Algorithms (B.Sc.) <u>Dr. A. Dolati</u> - Contributed to assignment-question design and evaluated student submissions.
Sep 2023–Jan 2024	Foundations of Numerical Analysis (B.Sc.) <u>Dr. M. Abbaszadeh</u> - Led problem-solving and tutorial sessions to review course material and address student questions. Introduction to the Theory of Computation (B.Sc.) <u>Dr. N. Roshandel Tavana</u> - Led problem-solving and tutorial sessions and evaluated student assignments.

Skills

Programming	Python, C++, Java, C, Rust
ML & Deep Learning	PyTorch, Hugging Face Transformers, scikit-learn
Scientific & Image Computing	NumPy, SciPy, OpenCV, scikit-image, pandas
Bioinformatics	Scanpy, scVelo, AlphaFold2, PyMOL
Tools	Git, GitHub, L ^A T _E X
Languages	English (CEFR C1); Persian (Native)

Volunteering

Jul 2026 – Present	Student Scientific Magazine “Halghe” Amirkabir University of Technology Contributed to scientific editing and editorial review, and suggested relevant scientific articles and books for publication and discussion.
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